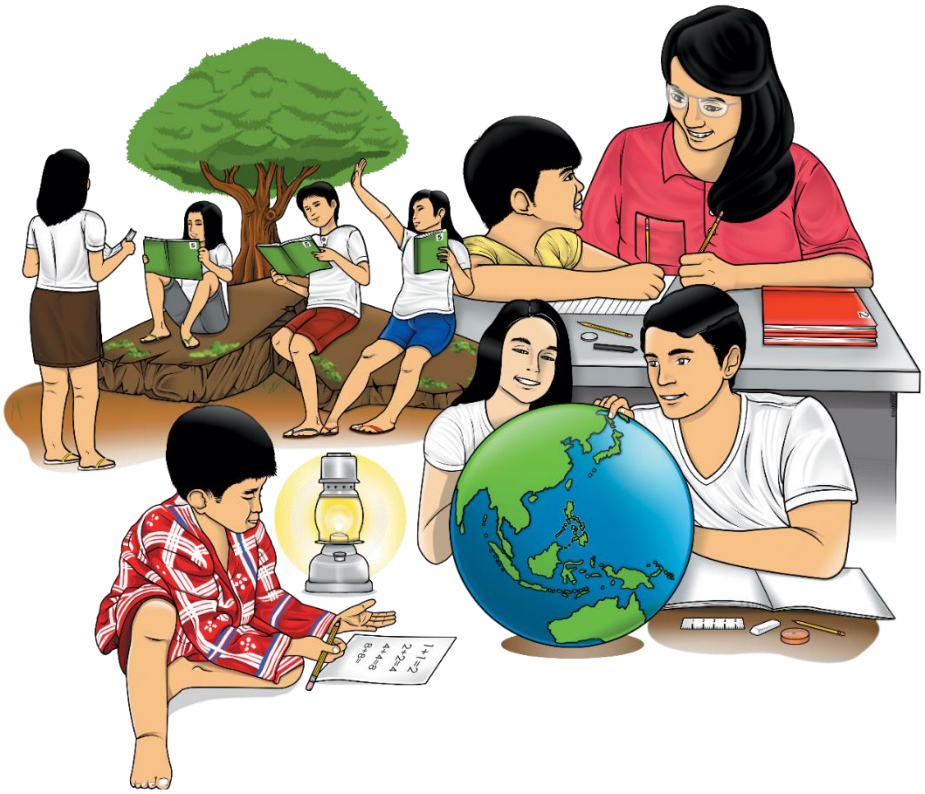
Senior High School

General Mathematics

Quarter 1: Rational, Exponential, and Logarithmic Functions



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Quarter 1: Rational, Exponential, and Logarithmic Functions
Second Edition, 2021

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Lesson 1	Key Concepts of Functions

This module was designed and written with you in mind. It is here to help you master the nature of Mathematics. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

What I Need to Know

After going through this module, you are expected to:

1. represent real-life situations using functions, including piecewise functions (M11GM-Ia-1);
2. evaluate a function (M11GM-Ia-2);
3. perform addition, subtraction, multiplication, division and composition of functions (M11GM-Ia-3);
4. solve problems involving functions (M11GM-Ia-4).

What I Know

Choose the letter of the best answer. Write the chosen letter on a separate sheet of paper.

1. When evaluated at $x = 3$, which of the following functions will result to 4?
   a. $f(x) = x^2 - 2$
   b. $f(x) = 2x^2 - 5x$
   c. $f(x) = -x^2 + 5$
   d. $f(x) = 2x^2 - 5x + 1$

For **nos. 2 – 3;** please refer to the following functions:

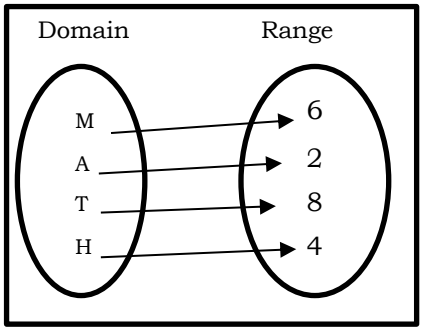
$$f(x) = x^2 - 3 \text{ and } g(x) = 2x + 5$$

2. What is the sum of the given functions?
   a. $(f + g)(x) = x^2 + 2x + 2$
   b. $(f + g) = x^2 + 2x - 2$
   c. $(f + g)(x) = x^2 - 2x + 2$
   d. $(f + g)(x) = x^2 + 2x - 2$
3. Which of the following operations must be performed for *f(x)* and *g(x)* to obtain an answer of $x^2 - 2x - 8$?
   a. $\frac{f(x)}{g(x)}$
   b. $f(x) \bullet g(x)$
   c. $(f - g)(x)$
   d. $f(g(x)$

What's In

Relation and Function

A **relation** is a pairing of input values with output value. It can be shown as a set of ordered pairs (x, y), where x is an input and y is an output. The set of input values x for a relation is called a **domain,** and the set of output values y is called the **range.**

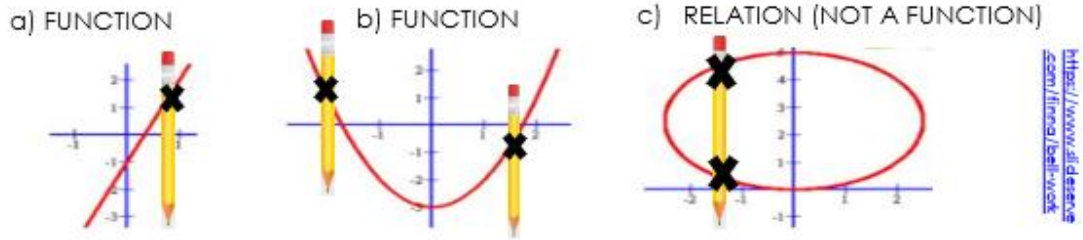


From the figure on the left, we can express the set of **ordered pairs** in the form {(M ,6), (A, 2), (T, 8), (H, 4)}. Observe that the first coordinate is different in each ordered pair. This relation shows a **function**. A **function** is a relation where there is only one output y for each input x. So, each element of the domain is mapped to exactly one element in the range.

Another way of representing a relation or function is through **table of values**. For example, the table on the right shows the cooking time of rice for each number of servings. The number of servings represent the domain {2, 4, 6, 8} while the cooking time in minutes represent the range {5, 8, 10, 11}. There is only one cooking time for each number of servings. Therefore, the relation from number of servings to cooking time is a function.

Cooking Time of Rice				
Servings	2	4	6	8
Cooking time in minutes	5	8	10	11

A function may also be represented by a graph. The **vertical line test,** also known as "pencil test", can be used to determine if the given relation is a function or just mere relation. For a graph to be considered a function, when a vertical line is drawn through the graph it must only pass through it at exactly one point.



From the figures above, graphs a and b are functions because each of our pencil test passes through the graph at only one point (represented by ✖). However, for graph c, the vertical line passes through the graph in more than one point which makes it a mere relation and therefore, not a function.

What's New

Evaluating Functions

Evaluating a function means substituting a given value of x in the function to find the value of f(x). Remember other variables may be used to represent a function other than f(x) such as g(x), c(h), w(r) and others.

Examples: Evaluate the following functions.

1. $k(r) = 2r + 1$ when $r = 9$ By replacing r with 9, the function will become, $k(9) = 2(9) + 1$ $= 18 + 1$ $\boldsymbol{k(9) = 19}$	3. $p(m) = \frac{2m+1}{m-1}$ when m = 2 By replacing m with 2, the function will become, $p(2) = \frac{2(2)+1}{2-1} = \frac{5}{1}$ $\boldsymbol{p(2) = 5}$

2. $f(x) = x^2 - 2x + 2 \; when \; x = 3$

By substituting $x = 3$, the function will become,

$$f(3) = (3)2 - 2(3) + 2$$
$$= 9 - 6 + 2$$
$$\boldsymbol{f(3) = 5}$$

4. $p(m) = \frac{2m+1}{m-1}$ when m = 2

By replacing m with 2, the function will become,

$$p(2) = \frac{2(2)+1}{2-1} = \frac{5}{1}$$
$$\boldsymbol{p(2) = 5}$$

Operations on Functions

Performing operations on functions is much the same way as performing operations on numbers or expressions. Add, subtract, multiply or divide functions by operating on their rules.

Notation for Operations on Functions	
Operation	**Notation**
Addition	(f + g)(x) = f(x) + g(x)
Subtraction	(f - g)(x) = f(x) - g(x)
Multiplication	(f•g)(x) = f(x) • g(x)
Division	$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, where\ g(x) \neq 0$

Examples:

Given $f(x) = 2x + 6$, and $g(x) = x + 3$, find:

1. $(f + g)(x) = f(x) + g(x)$
   $= (2x + 6) + (x + 3)$
   $\boldsymbol{(f + g)(x) = 3x + 9}$

Combine similar terms: 2x + x and 6 + 3

2. $(f - g)(x) = f(x) - g(x)$
   $= (2x + 6) - (x + 3)$
   $= 2x + 6 - x - 3$
   $\boldsymbol{(f - g)(x) = x + 3}$

Combine similar terms: 2x - x and 6 - 3

3. $f \cdot g)(x) = f(x) \cdot g(x)$
   $= (2x + 6)(x + 3)$
   $\boldsymbol{(f \cdot g)(x) = 2x^2 + 12x + 18}$

Multiply the factors. Recall FOIL method
First term: (2x)(x) = $2x^2$
Outer term: (2x)(3) = 6x
Inner term: (6)(x) = 6x
Last term: (6)(3) = 18
Then, simplify by combining similar terms.

4. $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$
   $= \frac{2x+6}{x+3}$
   $= \frac{2\cancel{(x+3)}}{\cancel{x+3}}$
   $\boldsymbol{\left(\frac{f}{g}\right)(x) = 2}$

In division, always check if expressions are factorable so you can arrive at the answer in simplest form.

In this case, using GCF, 2x + 6 can be factored as 2(x+3).

Composition of Functions

Composition of functions uses the output from one function as the input for a second function. The composition of functions f and g is denoted by $(f \circ g)(x) = f(g(x))$ it is read as *"f of g of x"* or *"f is composed of g of x"*.

Example:

Given $f(x) = x^2 - 3$ and $g(x) = 2x - 5$, find $(f \circ g)(4)$.

Solving for *g(4)*:
$g(x) = 2x - 5$
$g(4) = 2(4) - 5$
$= 8 - 5$
$g(4) = 3$
Substituting this result to $f(x)$, we have,
$f(x) = x^2 - 3$
$f(3) = (3)^2 - 3$
$= 9 - 3$
$f(3) = 6$
Thus, $\boldsymbol{(f \circ g)(4) = f(g(4)) = 6}$

What is It

Piecewise - defined Function

Suppose your mobile plan charges you ₱300 monthly. The plan includes 100 free text messages however, in excess of 100 text messages you will be charged ₱1 for every message sent. This is an example of a piecewise function.

A piecewise -defined function is a function that is defined on a sequence of intervals. Another way of representing a function is through equation. Thus, if we represent the number of text messages sent as *x,* the example above can be expressed as:

$$f(x) = \begin{cases} 300; & if\ 0 < x \leq 100 \\ 300 + (x - 100); & if\ x > 100 \end{cases}$$

This means that you only have to pay ₱300 if the number of text messages sent is between 0 to 100.

₱300 is charged if the number of messages you sent, x, *is* between 0 and 100 or $0 < x \leq 100$. However, if you exceeded the 100 free text messages you will be charged ₱1 for every excess text message.

Suppose the number of text messages is 170 then you have to pay ₱70 more on top of your ₱300 mobile plan. Using the second equation in the piecewise defined function since the number of text messages sent, x, is more than 100, that is, $x > 100$ then we will use the equation $f(x) = 300 + (x - 100)$. Since x = 170 the equation will become,

$$\begin{aligned} f(170) &= ₱\,300 + (170 - 100) \qquad (substitute\ x\ with\ 170) \\ &= ₱300 + 70 \\ f(170) &= ₱370 \end{aligned}$$

This means you have to pay ₱370 if you were able to consume 170 text messages in a month.

What's More

A jeepney ride costs ₱ 9.00 for the first 4 kilometers, and each additional kilometer adds ₱ 1.50 to the fare. Use the piecewise - defined function to represent the jeepney fare in terms of the distance ***d*** in kilometers.

What I Have Learned

Directions: Fill in the blanks with the correct word or phrase to complete the given statement.

1. A function can be represented in the following forms: _____.
2. The vertical line test must intersect the graph at _____ one point.
3. Evaluating a function *f(x)* means _____ a given value of x in the function to find the value of f(x).
4. _____ of function uses the output from one function as the input for a second function.
5. A Piecewise - Defined function is a function that is defined on a sequence of _____.

What I Can Do

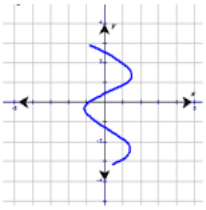
For your upcoming birthday party, you are planning to hire a catering service. The caterer charges ₱200 per head for 20 persons or less, ₱180 per head for 21 to 50 persons, and ₱160 per head for 51 to 100 persons.

1. Create the piecewise – defined function for this situation.
2. How much do you have to pay the caterer if 35 persons attended your party?

Assessment

I. Write YES if the following represents a function, write NO if it does not.

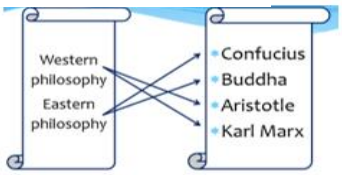
_____1.



_____ 3.

CODE	COUNTRY
Ph	Philippines
Gr	Germany
Jpn	Japan
UK	United kingdom

_____2.



_____4. {(0,0), (1,1),(2,4),(3,9),…(n,n^2),…}

_____5. {(teacher, school), (sales lady, store), (accountant, bank), (auditor, bank)}

II. Perform as what is indicated.

A. Evaluate the functions at $x = 3$.

1. $f(x) = x - 3$
2. $g(x) = x^2 - 3x + 5$
3. $h(x) = \sqrt[3]{x^3 + x + 3}$
4. $p(x) = \frac{x^2+1}{x-4}$

B. Use the function given in part A to perform the operations indicated below.

1. $f(3) + p(3)$
2. $p(5) - g(3)$
3. $p(3) \bullet g(3)$
4. $\frac{f(100)}{p(3)}$
5. $f(g(x))$
6. $f(p(5))$

Additional Activities

Suppose that $s(t)$ is the top speed (in km per hour) of a runner when the temperature is t degrees Celsius. Explain what the statements s(15) = 12 and $s(30) = 10$ mean.

Lesson 2	Key Concepts of Rational Functions, Equations and Inequalities.

What I Need to Know

After going through this module, you are expected to:

1. represent real – life situations using rational functions (M11GM-Ib-1);

2. distinguish rational function, rational equation, and rational inequality (M11GM-Ib-2);
3. solve rational equations and inequalities (M11GM-Ib-3);
4. represent a rational function through its: (a) table of values, (b) graph, and (c) equation (M11GM-Ib-4);
5. find the domain and range of a rational function (M11GM-Ib-5).

What I Know

Choose the letter of the best answer. Write the chosen letter on a separate sheet of paper.

_____1. Which of the following refers to an expression which can be written as a ratio of two polynomials?

a. Function
b. Rational Expression
c. Algebraic l Expression
d. Polynomial Expression

_____2. Which of the following is the solution for the rational equation $\frac{2}{x} - \frac{3}{2x} = \frac{1}{5}$?

a. $x = -\frac{5}{2}$
b. $x = -\frac{2}{5}$
c. $x = \frac{2}{5}$
d. $x = \frac{5}{2}$

_____3. Which of the following is/are the solution/s for the rational equation $\frac{x}{x+2} - \frac{1}{x-2} = \frac{8}{x^2-4}$?

a. $x = -2$
b. $x = 5$
c. $x = -2 \; or \; x = -5$
d. $x = -2 \; or \; x = 5$

What's In

Now that you have learned the key concepts of function namely evaluating functions, operations on function, composition of functions, piecewise defined function, and solving problems involving those concepts, let us move forward and keep learning more about functions. This time, we will explore rational functions. Let's get it on!

What's New

- ✓A **rational expression** is an expression which can be written as a ratio of two polynomials. Examples: $\frac{2}{x}$, $\frac{x^2+2x+3}{x+1}$; $\frac{5}{x-3}$.
- ✓A **rational equation** is an equation involving rational expressions which can be solved for all unknown values that satisfy the equation. Examples: $\frac{x+1}{3} = \frac{2}{3}$; $\frac{2}{x} - \frac{3}{2x} = \frac{1}{5}$
- ✓A **rational inequality** is an inequality involving rational expressions which can be solved for all unknown values satisfying the inequality. Examples: $\frac{x}{2} > \frac{9}{4}$; $\frac{5}{x-3} \leq \frac{2}{x}$
- ✓A **rational function** can be written in the form $f(x) = \frac{p(x)}{q(x)}$ where $p(x)$ and $q(x)$ are polynomials and $q(x)$ is not a zero polynomial that is, $q(x) \neq 0$.
  Examples: $f(x) = \frac{x-7}{x+9}$; $f(x) = \frac{x^2+2x+3}{x+1}$

What is It

Ways of Representing Rational Functions

A **rational function** can be written in the form $\boldsymbol{f(x)} = \frac{\boldsymbol{p(x)}}{\boldsymbol{q(x)}}$ where $p(x)$ and $q(x)$ are polynomials and $q(x)$ is not a zero polynomial that is, $q(x) \neq 0$. There are several ways to represent rational functions these are **using table of values, equations, and graphs**.

As an example, the average speed (or velocity) which can be computed by using the formula $s = \frac{d}{t}$, where s represents speed, d for distance and t is time, can be expressed in the form of rational function. We will use this concept of speed to discuss the ways of representing rational function.

✓ **Rational Function in Equation Form**

Example 1: Suppose you are a runner in a marathon, how would you represent the time it takes for you to run 100 meters in the track?

Solution: Since your speed depends on the time it takes you to run 100 meters, we can represent speed as a function of time. Let t represent the time it takes for you to run 100 meters. Then the speed can be represented as a function s(t) as $\boldsymbol{s(t)} = \frac{\boldsymbol{d}}{\boldsymbol{t}}$ since a distance of 100 is given then, $\boldsymbol{s(t)} = \frac{\boldsymbol{100}}{\boldsymbol{t}}$ **.**

✓ **Rational Function as Table of Values**

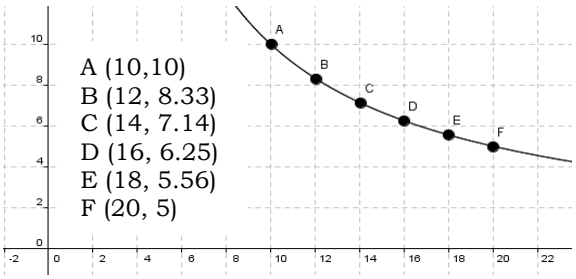
Example 2: Continuing the scenario above, construct a table of values for your speed against the given different run times as shown in the table below.

Solution: Using the function $\boldsymbol{s(t)} = \frac{\boldsymbol{100}}{\boldsymbol{t}}$, we will be able to determine your speed at different times by simply substituting the value of t in the function and performing the indicated operation. The table of values for your run time from 10 to 20 seconds is shown on the right.

t	10	12	14	16	18	20
$s(t)$	10	8.33	7.14	6.25	5.56	5

✓ **Rational Function as a Graph**

Example 3: From the table in example 2, we can observe that the speed decreases with time. By plotting the points on the Cartesian plane from each ordered pairs on the table of values, we can see that it follows a smooth curve.



Domain and Range of Rational Function

The **domain** of a rational function is the set of all real numbers that are ***not*** zeros of its denominators. To simply put it, these numbers do not make the denominator equal to zero. When the numbers from the domain are substituted in the function, the set of resulting values is called the **range.**

Example: Find the domain of $\boldsymbol{f(x)} = \frac{\boldsymbol{x^2 + 3x + 1}}{\boldsymbol{x^2 - x - 6}}$

Remember that domain in rational function deals with the denominator, so we will focus our solution on the denominator $x^2 - x - 6$ and equate it to zero.

Solution:

$x^2 - x - 6 = 0$

(x-3)(x+2)=0

x-3=0	x+2=0
$\mathbf{x = 3}$	$\mathbf{x = -2}$

The domain is the set of all real numbers except $x = 3$ or $x = -2$, because the denominator is 0 when $x = 3$ or $x = -2$, making the fraction undefined. Thus, **Domain:** $\{x \in R | x \neq 3 \text{ or } x \neq -2\}$.

Solving Rational Equations

In solving rational equations, we are solving for the value of the variable that will satisfy the given equation. To simply put it, we are looking for all possible value(s) that will make the left side of the equation equal to the right side.

To solve a rational equation:

1. First, eliminate denominators by multiplying each term of the equation with the least common denominator (LCD);
2. Then, check the solution(s) using the original equation. Remember that it is possible to arrive at extraneous solutions, which means that the values do not satisfy the condition of the original equation. Thus, it is essential that solutions are checked.

Example 1: Solve $\frac{6}{x} + \frac{5}{4} = -\frac{7}{4}$

The LCD is $\mathbf{4x}$. Multiply each term of the equation with the LCD

$\mathbf{4}\cancel{\mathbf{x}}\left(\frac{6}{\cancel{x}}\right) + \left(\frac{5}{\cancel{4}}\right)\cancel{\mathbf{4}}\mathbf{x} = \left(-\frac{7}{\cancel{4}}\right)\cancel{\mathbf{4}}\mathbf{x}$ *multiply the remaining factors of each term*

$4(6) + 5x = -7x$

$24 + 5x = -7x$

$5x + 7x = -24$ *combine similar terms by applying addition property of equality then simplify*

$\frac{\cancel{12}x}{12} = \frac{-24}{12}$

$\mathbf{x = -2}$

Try and check if $x = -2$ is a solution to the original equation. The following guide questions will help you:

- ✓ Does the value $x = -2$ satisfy $\frac{6}{x} + \frac{5}{4} = -\frac{7}{4}$?
- ✓ Is the left side of the equation equal to the right?

If both your answer is yes then, you are correct! x = -2 is a solution to the given rational equation.

Example 2: Solve $\frac{1}{x-2} = \frac{3}{x+2} - \frac{6x}{x^2-4}$

The LCD is $x^2 - 4$ or $(x+2)(x-2)$. Multiply each term by this LCD.

$(\mathbf{x+2})\cancel{(\mathbf{x-2})}\left(\frac{1}{\cancel{x-2}}\right) = \left(\frac{3}{\cancel{x+2}}\right)\cancel{(\mathbf{x+2})}(\mathbf{x-2}) - \left(\frac{6x}{\cancel{x^2-4}}\right)\cancel{(\mathbf{x+2})(\mathbf{x-2})}$

$(x+2)(1) = 3(x-2) - 6x$ Multiply the remaining factors

$x + 2 = 3x - 6 - 6x$

$x + 2 = -3x - 6$ Apply the addition property of equality (APE) and combine similar terms.

$x + 3x = -6 - 2$

$$\frac{\cancel{4}x}{\cancel{4}} = -\frac{8}{4}$$
$$\boldsymbol{x = -2}$$

If $x = -2$ is substituted in the original equation, a denominator of zero will be obtained. Remember, division by zero is undefined. So, $x = -2$ is an **extraneous solution**, and the original equation has **no solution.**

Solving Rational Inequalities

The inequality symbols < , > , ≤ , and ≥ are part of rational inequalities. Like rational equations, we can also solve rational inequalities.

Example: Solve the inequality $\frac{2x}{x+1} \geq 1$.

Solution:

Step 1*: Rewrite the inequality as a single rational expression with zero on one side.*

$$\frac{2x}{x+1} \geq 1$$

Add -1 to both sides of the inequality

$$\frac{2x}{x+1} - 1 \geq 0$$

Make the terms similar by multiplying each term on the left side of the inequality by the LCD.

$$\frac{2x - (x+1)}{x+1} \geq 0$$

Simplify by combining similar terms.

$$\frac{2x - x - 1}{x+1} \geq 0$$

$$\boldsymbol{\frac{x-1}{x+1} \geq 0}$$

Step 2: *Try some values or test points for x in the inequality* $\frac{x-1}{x+1} \geq 0$

Test points may be determined by equating the numerator and denominator to zero and solving for x.

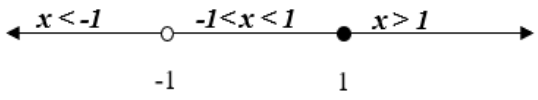
Numerator	Denominator
$x - 1 = 0$	$x + 1 = 0$
$\boldsymbol{x = 1}$	$\boldsymbol{x = -1}$

a) Substitute these test points in the inequality based on the answer in step 1:

Test point x = 1 $\frac{1-1}{1+1} \geq 0$ $\frac{0}{2} \geq 0$	The rational expression is zero when x =1 and it satisfies the given inequality. Thus, the value **x =1 is included in the solution.**	**Test point x = -1** $\frac{-1-1}{-1+1} \geq 0$ $\frac{-2}{0} \geq 0$ ***Undefined***	The rational expression is undefined at x = -1. Thus, **x = -1 is NOT included in the solution.**

b) then represent on the number line

To represent this on the number line, use **shaded circle** for $x = 1$ to indicate that it is **included in the solution** and use **unshaded circle** for $x = -1$ which means that it is **not included in the solution.**



Step 3*: Choose other test points in the intervals* determined by -1 and 1 to determine the sign of $\frac{x-1}{x+1}$ in these intervals. For convenience, we will construct a table of signs as shown below. To complete the table of signs simply substitute the value of the test points in both the numerator and denominator to determine the resulting signs.

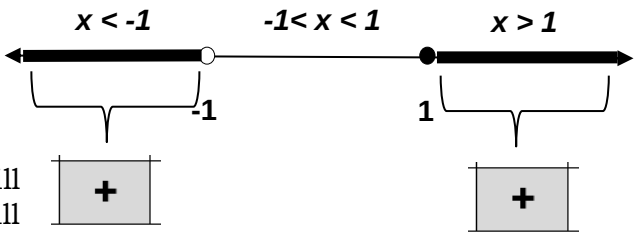
For example, in test point $x = 0$

- The *numerator* $x - 1$ will become 0 -1 = - 1. Thus, the resulting sign is **negative.**
- The *denominator* $x + 1$ will become 0 + 1 = 1. Thus, the resulting sign is **positive.**
- When these two signs are expressed in rational form the resulting sign is **negative** since $\frac{negative}{positive} = \frac{-}{+} = -$

Interval	x < -1	-1< x < 1	x > 1
Test Point	x = -2	x = 0	x = 2
Numerator: $x - 1$	-	-	+
Denominator: $x + 1$	-	+	+
$\frac{x-1}{x+1}$	**+**	**-**	**+**

Step 4*. Look at the inequality symbol in the given problem to identify the solutions needed.*

According to the inequality $\frac{x-1}{x+1} \geq 0$ we are looking for the intervals where the rational expression is **positive** (due to the symbol greater than) **or zero**. Thus, by plotting this on the number line we will be able to see the solutions that will satisfy the inequality.



Therefore, the solution for the rational inequality $\frac{x-1}{x+1} \geq 0$ is any real number less than -1 or greater than or equal to 1. In set builder notation the solution is $\{\boldsymbol{x} \in \mathbb{R} | \boldsymbol{x} < -\mathbf{1}\ \boldsymbol{or}\ \boldsymbol{x} \geq \mathbf{1}\}$.

What's More

Solve the following rational equation/inequality.

1. $\frac{2x-9}{x-7} + \frac{x}{2} = \frac{5}{x-7}$
2. $\frac{10}{x+8} > 2$ (express your answer in set notation)

What I Have Learned

Directions: Fill in the blanks with the correct word or phrase to complete the given statement.

1. A _____can be written in the form $\boldsymbol{f(x)} = \frac{\boldsymbol{p(x)}}{\boldsymbol{q(x)}}$ where $p(x)$ and $q(x)$ are polynomials and $q(x)$ is not a zero polynomial that is, $q(x) \neq 0$.
2. Rational functions can be represented through_____.
3. The zeroes of a rational function is the set of all real numbers that are ***not*** zeroes of its _____. When the numbers from the domain are substituted in the function the set of resulting values is called the _____.

What I Can Do

Solve the following rational equation/inequality.

1. $\frac{16}{x^2-16} = \frac{2}{x-4}$
2. $\frac{6}{x-2} \geq -4$ (express your answer in set notation)

Assessment

I. Identify if the following is a **rational function, rational equation, rational inequality** or **none of these**.

_____1. $\frac{2+x}{x-7} = 9$

_____2. $x > \sqrt{3x+2}$

_____3. $f(x) = 6 - \frac{x+2}{x^2-8}$

_____4. $2x \geq \frac{8}{3x+5} = 8$

_____5. $\frac{x}{3} = \frac{9}{x+5x^2}$

II. Solve for the value of the variable in the rational equation $\frac{10}{3} = \frac{4}{x} + 3$. Remember to perform solution check to verify if your answer satisfies the given equation.

III. Solve for x in the inequality $\frac{4}{x+1} < 4$. Express your answer in set notation.

Additional Activities

Solve the inequality $\frac{4}{2x-1} \geq \frac{1}{x+1}$ and express the answer in set notation.

Lesson 3 — Key Concepts of Rational Functions: Intercepts, Asymptotes, Zeroes and Problem Solving

What I Need to Know

After going through this module, you are expected to:

1. determine the intercepts, zeroes, and asymptotes of rational functions (M11GM-Ic-1);
2. solve problems involving rational functions, equations, and inequalities (M11GM-Ic-3).

What I Know

Choose the letter of the best answer. Write the chosen letter on a separate sheet of paper.

_____1. A rational function can be written in the form of $f(x) = \frac{p(x)}{q(x)}$ with what restrictions?

a. $p(x)$ and $q(x)$ are not zero polynomials.

b. (x) and $q(x)$ are polynomials and $q(x) \neq 0$.

c. $p(x)$ and $q(x)$ are polynomials and $p(x) \neq 0$.

$d.$ $p(x)$ and $q(x)$ are polynomials and $q(x) = 0$.

_____2. Which of the following equation will determine the amount of time it will take to finish painting the house if Joey and Chandler work together?

a. $\frac{1}{4} + \frac{1}{3} = \frac{1}{x}$ b. $\frac{1}{4} - \frac{1}{3} = \frac{1}{x}$ c. $\frac{x}{4} + \frac{x}{3} = \frac{1}{x}$ d. $\frac{1}{4} + \frac{1}{3} = x$

_____3. How long would it take Joey and Chandler to finish painting the house?

a. ½ hour b. 1.71 hours c. 2.5 hours d. 3.75 hours

What's In

Rational Function

A **rational function** can be written in the form $f(x) = \frac{p(x)}{q(x)}$ where $p(x)$ and $q(x)$ are polynomials and $q(x)$is not a zero polynomial that is, $q(x) \neq 0$. There are also other terms that tells so much about a rational function, these are **x and y – intercepts, zeroes,** and **asymptotes.**

What's New

Intercept is the point, along the x or y – axes, a graph passes through.

- **y – intercept** is the point where the graph intersects the y-axis or where x=0.
- **x – intercept** is the point where the graph intersects the x-axis or where y=0.

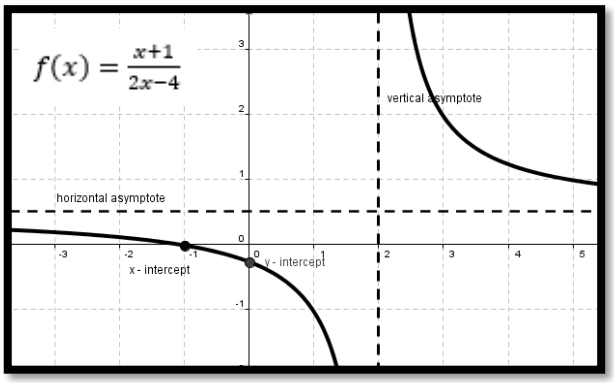
Asymptote is a line that approaches a given graph but does not meet at any finite distance. It simply tells that the graph gets closer and closer to that asymptote but will not actually cross, reach nor touch it because the function is undefined at the value of that asymptote. It is represented by broken lines for visualization.

- **Vertical asymptote** is a vertical line which correspond to the zeroes of the denominator.
- **Horizontal asymptote** is a horizontal line the graph approaches, it is concerned with the degree of both the numerator and denominator. Suppose $\boldsymbol{n}$ is the degree of the numerator and $\boldsymbol{m}$ is the degree of the denominator, then a comparison is made to identify the horizontal asymptote:
  - If $\boldsymbol{n} < \boldsymbol{m}$, the horizontal asymptote is $\boldsymbol{y} = \boldsymbol{0}$
  - If $\boldsymbol{n} = \boldsymbol{m}$, the horizontal asymptote is $\boldsymbol{y} = \frac{\boldsymbol{a}}{\boldsymbol{b}}$, where a is the leading coefficient of the numerator and b is the leading coefficient of the denominator.

- If $\boldsymbol{n} > \boldsymbol{m}$, there is **no horizontal asymptote.**

For us to easily determine the intercepts and asymptotes, let us follow the guide below using the function $f(x) = \frac{x+1}{2x-4}$.

y – intercept	**x – intercept**	**Vertical asymptote**	**Horizontal asymptote**
set x = 0 $f(x) = \frac{x+1}{2x-4}$ $f(0) = \frac{0+1}{2(0)-4}$ $= \frac{1}{0-4}$ $\boldsymbol{f(0) = -\frac{1}{4}}$ Therefore, the y – intercept is $\left(0, -\frac{1}{4}\right)$.	Equate the numerator to zero then, solve for x. $x + 1 = 0$ $x = -1$ Therefore, the x – intercept or zero(s) is **(-1,0)**.	Equate the denominator to zero, then solve for x. $2x - 4 = 0$ $2x = 4$ $\frac{\cancel{2}x}{\cancel{2}} = \frac{4}{2}$ $\boldsymbol{x = 2}$ 2 is not a zero of the numerator (compared from the previous column).Thus, the vertical asymptote is the line **x = 2.**	*Numerator:* ***x+1*** ➢ *x* in the numerator *means* x^1 so the degree is **1,** let's call it **n.** ➢ The leading coefficient of x is 1 because *x* also means 1x, let's call it **a.** *Denominator:* ***2x-4*** ➢ *2x* in the denominator means $2x^1$, so degree is also **1,** let's call it **m.** ➢ The leading coefficient is 2 from *2x, let's call it* ***b****.* When n and m are compared, which makes it true, n < m, n = m, n > m? Obviously, it is n = m since 1 = 1 therefore the horizontal asymptote is determined by $y = \frac{a}{b} = \frac{1}{2}$ Therefore, our horizontal asymptote is the line $\boldsymbol{y = \frac{1}{2}}$.



What is It

Solving Work Problems

The amount of work done is equal to the product of the rate at which work is being done and the amount of time required to do the work. That is, **Work = Rate x Time.**

Example 1: Suppose that Mang Carding can plow a rice field at a rate of 3 fields per hour. How many rice fields can he plow in 6 hours?

Solution: $Work = 3\ \frac{rice\ fields}{hour} x\ 6\ hrs = 18\ rice\ fields.$

Example 2: It takes Benie 6 hours to paint a house and June takes 5 hours to paint the same sized house. How long will it take them to paint one house of the same size together?

Solution: Benie can paint 1 house in 6 hours means Benie can paint at a rate of $\frac{1}{6}$ of the house per hour while June can paint 1 house in 5 hours means June can paint at a rate of $\frac{1}{5}$ of the house per hour. Their combined rate is $\frac{1}{x}$ thus we will have the equation,

Benie's rate

June's rate

Rate when they work together

$$\frac{1}{6} + \frac{1}{5} = \frac{1}{x}$$

$30x\left(\frac{1}{6}\right) + \left(\frac{1}{5}\right)30x = \left(\frac{1}{x}\right)30x$ multiply each term by the LCD $30x$

$\frac{30x}{6} + \frac{30x}{5} = \frac{30x}{x}$ simplify each term

$5x + 6x = 30$ combine similar terms

$11x = 30$

$\frac{\cancel{11}x}{\cancel{11}} = \frac{30}{11}$ simplify to solve for x

$x = \frac{30}{11}$

$x = 2.73\ hours$ or
$2\ hours, 43\ minutes\ and\ 48\ seconds$

Therefore, it will take Benie and June 2.73 hours or 2 hours, 43 minutes and 48 seconds hours to finish painting a house together.

What's More

Suppose you are tasked in school to finish a report. You know that you can accomplish it in 4 days. Your friend offered to help you because she knows she can finish the same task in 2 days. How long will it take to finish the report if you and your friend work together?

What I Have Learned

Directions: Fill in the blanks with the correct word or phrase to complete the given statement.

1. _____simply tell at what point, along the x or y – axes, a graph passes through.
2. _____ are represented by a broken line which tells that the graph gets closer and closer to it but will not actually cross nor touch the line.
3. The amount of work done is equal to the product of the _____ at which work is being done and the amount of _____required to do the work.

What I Can Do

1. It takes Andy 40 minutes to do a particular job alone. It takes Brenda 50 minutes to do the same job alone. How long would it take them if they worked together?
2. Identify the intercepts, asymptotes, and sketch the graph of $f(x) = \frac{x^2-x-2}{x-1}$.

Assessment

A. Solve for the x and y – intercepts, vertical and horizontal asymptotes of the following rational functions.

1) $f(x) = \frac{3x-6}{5-2x}$ 2. $f(x) = \frac{x^2-3x-10}{x}$

B. The triplet brothers named Justin, Jason, and Jacob are working on a school project. Justin can complete the project by himself in 6 hours, Jason can complete the project by himself in 9 hours, and Jacob can complete the project by himself in 8 hours. How long would it take the triplets to complete the project if they work together?

Additional Activities

One pipe can fill a swimming pool in 10 hours, while another pipe can empty the pool in 15 hours. How long would it take to fill the pool if both pipes were accidentally left open?

Lesson 4	Key Concepts of Inverse Function

What I Need to Know

After going through this module, you are expected to:

1. represent real – life situations using one-to-one functions (M11GM-Ic-3);
2. determine the inverse of a one-to-one function (M11GM-Id-2);
3. represent an inverse function through it's a) table of values and b) graph (M11GM-Id-3);
4. find the domain and range of an inverse function (M11GM-Id-4).

What I Know

Choose the letter of the best answer. Write the chosen letter on a separate sheet of paper.

_____1. Which of the following is used to test if the graph of the function is one – to – one?

a. line y=x b. vertical line c. horizontal line d. diagonal line

_____2. What happens with the domain and range when the inverse of a function is taken?

a. The domain and the range interchange.
b. The domain and the range remain the same.
c. The domain and the range increased by a value of 1.
d. The domain and the range decrease by a value of 1.

What's In

Your Learner's Reference Number or LRN is unique. Can you imagine when your LRN is encoded in the computer and a different name of student comes up? That's how important these numbers are. Your LRN is just one of the many examples of a one – to – one relationship. Other situations with one – to – one relationship are: SSS member to its Social Security Number, SIM cards to cellphone numbers, Bank account number to its owner, airport codes, etc.

What's New

One-to-One Function

A function f is one-to-one if $f(a) = f(b)$, which implies that $a = b$. It also means that the same y-value is never paired with two different x -values.

Examples:

1. {(0,0), (1,1), (2,8), (3,27), (4,64)} is a one-to-one function since no y – value is paired with two different x – values. Or simply, the y – value is only paired with one x – value.

2. {(-2,4), (-1,1), (0,0), (1,1), (2,4)} is **not** one-to-one function since there are y – values that are paired up with two different x – values. Specifically, the y-value 1 is paired with x values -1 and 1 in (-1,1) and (1,1). Also, the y – value 4 is paired with x – values -2 and 2 in (-2,4) and (2,4).

3. Consider an element of an electrical circuit which increases its resistance r whereas a function of temperature t also increases.
On the other hand, if we wish to determine the temperature t as a function of the resistance r, say t = g(r) then, we can say that g(r) and f(t) are inverse functions.

Let r*(x)* = *f(t)* thus,

Temperature	-20	0	20	40
Resistance	50	150	250	350

Resistance	50	150	250	350
Temperature	-20	0	20	40

What is It

The Inverse of One-to-One Function

A function has an inverse if and only if it is one-to-one. The importance of one-to-one functions is due to the fact that these are the only functions that have an inverse.

Definition: Let f be a one-to-one function with domain A and range B. Then the inverse of f, denoted by f^{-1}, is a function with domain B and range A. defined by: $f^{-1}(y) = x$ if and only if $f(x) = y$ for any y in B.

In the previous example if $r = f(t)$ where resistance is a function of temperature, then $t = f^{-1}(r)$. Temperature is the inverse function of resistance. It is read as *"the inverse of r".*

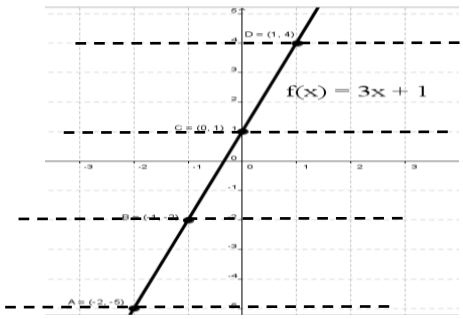
There are three conditions to satisfy if the function is one-to-one:

1. A function $y = f(x)$ with domain D is one-to-one on D if and only if for every x_1 and x_2 in D, $f(x_1) = f(x_2)$ implies that $x_1 = x_2$.
2. Each element y *or* $f(x)$ of the range is mapped from one and only one element x of the domain.
3. A function is one-to-one if and only if no horizontal line intersects the graph of $y = f(x)$ in more than one point.

Example: Identify the domain and range of $f(x) = 3x + 1$ and graph the function.

Solution: Since the given is a linear function, then the domain and range are the set of all real numbers. Domain: $\{x \mid x \in \mathbb{R}\}$ and Range: $\{y \mid y \in \mathbb{R}\}$

The graph of $f(x) = 3x + 1$ is shown below:



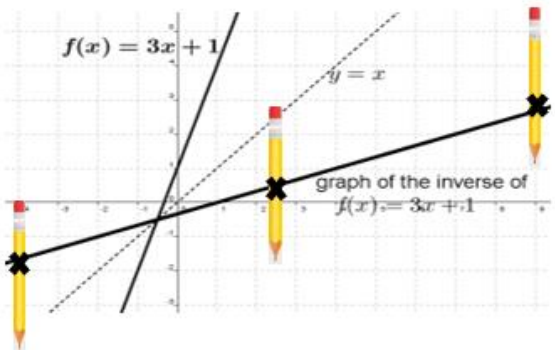
By reflecting along the line $y = x$ the two functions are **symmetrical** around that line (refer to the figure on the left).

The graph further shows that it satisfies the vertical line test which means that the inverse of $f(x) = 3x + 1$ is indeed a function.

To graph, we simply assign values of x and substitute it in the function $f(x) = 3x + 1$ to obtain the value of y *or* $f(x)$. Suppose we will use -2, -1, 0, 1, and 2 for x. Thus, the y values will be -5, -2, 1, 4, and 7. Refer to the table on the right.

x	-2	-1	0	1	2
$f(x)$	-5	-2	1	4	7

Through graphing, we can already determine the graph of the inverse of $f(x) = 3x + 1$ by simply reflecting its graph along the line $y = x$.



To determine if the graph of the function is one-to-one, we simply use **the horizontal line test**. To do this, we simply draw (or imagine) a horizontal line (using broken lines) anywhere on the graph of a function. For the function to be considered o one-to-one, the horizontal line must intersect the graph of the function once.

As shown on the graph, there are several horizontal broken lines and each of these only passes through the graph once. Therefore, the function $f(x) = 3x + 1$ is one-to-one.

If the domain and range of the inverse of the function is taken based from the table from values, we will have its corresponding table of values as shown on the right:

x	-5	-2	1	4	7
$f^{-1}(x)$	-2	-1	0	1	2

What's More

What happens with the domain and range of the function when its inverse is obtained?

What I Have Learned

Directions: Fill in the blanks with the correct word or phrase to complete the given statement.

1. A function has an inverse if and only if it is _____.
2. A function is one-to-one if and only if the horizontal line intersects the graph of $y = f(x)$ in _____ point.
3. Reflect the graph of a function along the line _____to determine the graph of its inverse.

What I Can Do

Given the function $f(x) = \frac{x-1}{3}$, do the following:

1. Construct the table of values by assigning values for x.
2. Based on the table of values, construct the graph of $f(x) = \frac{x-1}{3}$.
3. Compare your answers with the answers from our example. What do you observe?

Assessment

I. Write **true** if the statement is correct otherwise, write **false**.

_____1.All functions have their respective inverses which is also a function.

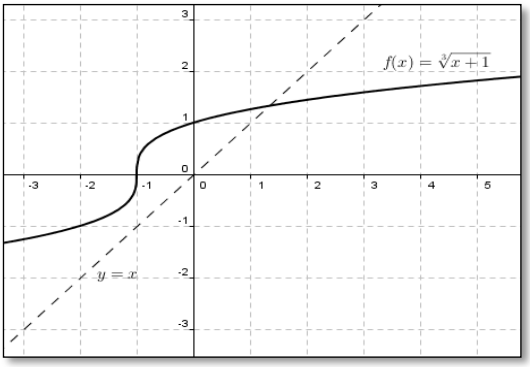
_____2. If a function has an inverse that is also a function, then it is a one-to-one function.

_____3. The vertical line test is used to determine if a graph represents a function while the horizontal line test is used to determine if the graph represents a one-to-one function.

_____4. A one-to-one function has y values paired to more than one x values.

_____5. In an inverse function, the domain becomes the range and the range becomes the domain.

II. Reflect the graph of $f(x) = \sqrt[3]{x+1}$ over the line $y = x$ using the given graph and answer the questions below.



1. Is the function $f(x) = \sqrt[3]{x+1}$ one – to – one? Why?
2. Is the inverse of $f(x) = \sqrt[3]{x+1}$ a function? Why?

Additional Activities

The equation $°F = \frac{9}{5}°C + 32$ can be used to convert between Celsius and Fahrenheit temperatures. What temperature is same in both the units?

Lesson 5	Inverse Functions and Exponential Functions

What I Need to Know

After going through this module, you are expected to:

1. solve problems involving inverse functions (M11GM-Ie-2);
2. represent real – life situations using exponential functions (M11GM-Ie-3);
3. distinguish between exponential function, exponential equation, and exponential inequality (M11GM-Ie-4).

What I Know

Choose the letter of the correct answer. Write the chosen letter on a separate sheet of paper.

1. If we solve for y in terms of x in 7x – y = 14, what is the resulting equation?
   a. y = 14 – 7x
   b. y = 7x – 14
   c. x = 14 + y
   d. $x = \frac{1}{2}y + 2$

2. A certain bacterium doubles every 30 minutes. If there were initially 100 bacterial cells to be cultured, how many bacteria will there be after 3 hours?
   a. 6,400 bacteria
   b. 800 bacteria
   c. 640 bacteria
   d. 8,000 bacteria

3. Suppose you are planning to save your money. You plan to save ₱1 for the first day, ₱2 for the second day, ₱4 for the third day, ₱8 for the fourth day, ₱16 on the 5th day and so on. If the pattern continues, how much money would you have saved for the 10th day?
   a. ₱1,024
   b. ₱512
   c. ₱200
   d. ₱2,000

What's In

Suppose you are considering two options to save up for your allowance. First, save a fixed amount of 10 pesos every day; or second, save starting with 5 pesos and double the amount every day. Which option would yield higher savings? The first option shows a linear increase at a fixed price of P10 per day while the second option produces higher savings as the amount is doubled every day. This shows an example of exponential growth.

What's New

Overview of Exponential Function, Exponential Equation, and Exponential Inequalities

✓ An **exponential function with base b** is a function of the form $f(x) = b^x$ *or* $y = b^x$
where $b > 0, b \neq 1$. Example: $f(x) = 0.5^x$ or $y = 0.5^x$
✓ An **exponential equation** is an equation involving exponential expressions.
Example: $7^{2x-x^3} = \frac{1}{343}$
✓ An **exponential inequality** is an inequality involving exponential expressions.
Example: $5^{2x} - 5^{x+1} \leq 0$

Exponential functions in real life

Example: At $t = 0$, there were initially 20 bacteria in a petri dish (a dish used to culture cells such as bacteria, fungi or small mosses). Suppose the time t *a* bacteria doubles every 100 hours. Give an exponential model for the bacteria as a function of t.

Solution: Initially, at

The bacteria doubles

$t = 0$	Number of bacteria = 20	starting day
$t = 100$	Number of bacteria = 20(2)	1st 100 hours
$t = 200$	Number of bacteria = $20(2)^2$	2nd 100 hours

If the pattern continues, an exponential model for this situation is $y = 20(2)^{\frac{t}{100}}$

What is It

Inverse Function

In solving problems involving inverse function, find the inverse by solving for one of the other variables.

Solving Problems Involving Inverse Function

Example: Gadgets in a store are sold at ₱4,999 each. If you pay by credit card a service charge of ₱75 will be added to your receipt. a) Write a function that gives the amount billed to the credit card c as a function of the number n of gadgets purchased and b) Write the inverse function from your answer in a.

Solution:
a) Think of the problem this way, your **credit card bill c** will be charged by **₱4,999 times a number n** depending on the number of gadgets you bought **plus a service charge of ₱75** since you will be using your credit card. Therefore, clearly we have $\boldsymbol{c = 4999n + 75}$.

b) Solving for the inverse means solving for the other variable which is n. So,

$$c = 4999n + 75$$

add -75 to both sides of the equation

$$c - 75 = 4999n$$

$$\frac{c-75}{4999} = \frac{\cancel{4999}n}{\cancel{4999}}$$

multiply both sides by $\frac{1}{4999}$ or divide both sides by 4999 then simplify.

$$\boxed{\frac{c-75}{4999} = n \text{ or } n = \frac{c-75}{4999}}$$

What's More

The number of insects triples every 5 days. If there are 100 insects to begin with, how many insects will there be on the 15th day?

What I Have Learned

1. An _____ with base b is a function of the form $f(x) = b^x$ *or* $y = b^x$
2. An _____ is an equation involving exponential expressions.
3. An _____ is an inequality involving exponential expressions.
4. In finding the _____, simply solve for one of the other variables.

What I Can Do

The equation $°F = \frac{9}{5}°C + 32$ can be used to convert between Celsius and Fahrenheit temperatures.

1. Express the equation in terms of degree Celsius.
2. The temperature for fever in adults starts at 37.6°C. What is the equivalent temperature in Fahrenheit?
3. What is the equivalent of $-40°F$ in Celsius?

Assessment

I. Identify the following if it is in the form of Exponential Function, Exponential Equation, or Exponential Inequality, or none of these.

_____1. $f(x) = 5x^2$

_____2. $2 \geq \left(\frac{1}{2}\right)^x$

_____3. $7^{4x} = y$ _____5. $y = 1.25^x$

_____4. $4(10^{x-2}) = 500$

II. From our example in *What is It*, we will use our answer $n = \frac{C-75}{4999}$ **:**

1. Find the number of gadgets ***n*** purchased when the credit card bill is ₱40,067.
2. Is it possible to have a total of ₱12,572.50 billed to your credit card for these gadgets? Why or Why not?

III. Create a model for the bacteria y if there were initially 5 bacteria in a petri dish and it doubles every 3 days.

Additional Activities

A salesclerk needs to price a product returned by a customer. The customer paid a total of ₱853.20, which included a gift – wrapping charge of ₱10 and 8% sales tax. What price should the clerk put back in the returned product?

Lesson 6	**Graphing Exponential Functions and Transformations of Exponential Functions**

What I Need to Know

After going through this module, you are expected to:

1. solve exponential equations and inequalities (M11GM-Ie-f-1);
2. represent an exponential function through its: (a) table of values, (b) graph, and (c) equation (M11GM-If-2);
3. find the domain and range of an exponential function (M11GM-If-3);
4. determine the intercepts, zeroes, and asymptotes of an exponential function (M11GM-If-4)

What I Know

Choose the letter of the correct answer. Write the chosen letter on a separate sheet of paper.

_____1. The following are the properties of exponential functions EXCEPT?

A. The domain is the set of all real numbers.
B. The y-intercept is 1. There is no x-intercept.
C. The range is the set of all positive real numbers.
D. It is a one-to-one function. It satisfies the Horizontal Line Test.

_____2. What is the horizontal asymptote of the function f(x) = 2^x?

A. y = 0 C. y = 2
B. y = 1 D. No horizontal asymptote

_____3. What is the vertical asymptote of the function f(x) = 2^x?

A. x = 0 C. x = 3
B. x = 1 D. No vertical asymptote

_____4.) For which value of the base (b) satisfies an increasing exponential function?

a. b > 1
b. b < 1
c. b ≤ 1
d. b ≠ 1

_____5.) For which value of the base (b) satisfies a decreasing exponential?

A. 0 > b < 1
B. 0 < b > 1.
C. 0 < b < 1.
D. None of these

What's In

Each output value is the product of the previous output and the base, 2. We call the base 2 the *constant ratio*. In fact, for any exponential function with the form f(x) = ab^x, b is the constant ratio of the function. This means that as the input increases by 1, the output value will be the product of the base and the previous output, regardless of the value of a.

What's New

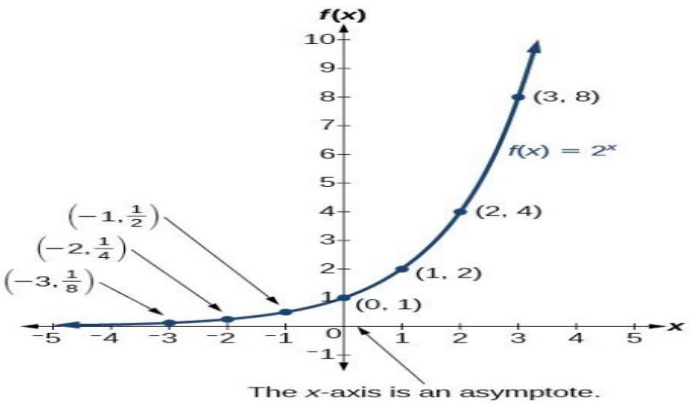
PROPERTIES OF EXPONENTIAL FUNCTIONS:

1. The domain is the set of all real numbers.
2. The range is the set of all positive real numbers.
3. It is a one-to-one function. It satisfies the Horizontal Line Test.
4. The y-intercept is (0, 1). There is no x-intercept.
5. The horizontal asymptote is the line y = 0 (or the x-axis). There is no vertical asymptote.
6. The function is increasing if b > 1, and is decreasing if 0 < b < 1.

$f(x) = 2^x$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

What is It

The graph below shows the exponential growth function f(x) = 2^x



https://courses.lumenlearning.com/wmopen-collegealgebra/chapter/introduction-graphs-of-exponential-functions/

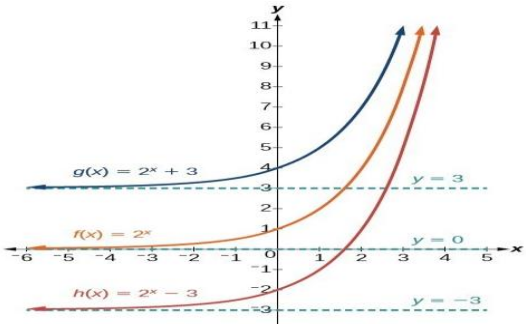
Notice that the graph gets close to the x-axis but never touches it.

The domain of f(x) = 2^x is all real numbers, the range is (0,∞), and the horizontal asymptote is y=0.

What's More

Graphing a Vertical Shift

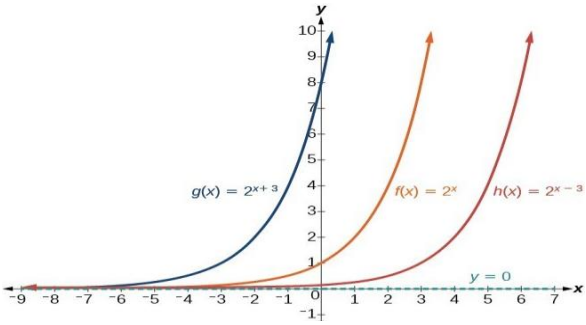
The first transformation occurs when we add a constant d to the parent function f(x) = b^x giving us a **vertical shift** d units in the same direction as the sign. For example, if we begin by graphing a parent function, f(x) = 2^x , we can then graph two vertical shifts alongside it using d=3: the upward shift, g(x) = 2^x + 3 and the downward shift, h(x) = $2^x - 3$. Both vertical shifts are shown in the figure on the right.



Department of Education General Mathematics Learner's Material

Graphing a Horizontal Shift

The next transformation occurs when we add a constant c to the input of the parent function f(x)=b^x giving us a **horizontal shift** c units in the *opposite* direction of the sign. For example, if we begin by graphing the parent function f(x)=2x, we can then graph two horizontal shifts alongside it using c=3c=3: the shift left, g(x)=2x+3, and the shift right, h(x)=2x−3. Both horizontal shifts are shown in the graph on the right.



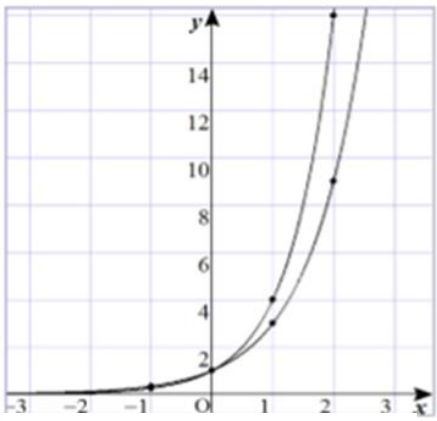
Department of Education General Mathematics Learner's Material

What I Have Learned

Graph the functions f(x) = 3^x and f(x) = 4^x in the same coordinate plane. Indicate the domain, range, y-intercept, and horizontal asymptote. Compare the two graphs.

Solution. For both these functions, the base is greater than 1. Thus, both functions are increasing. The following table of values will help complete the sketch.

x	-2	-1	0	1
f(x)	$\frac{1}{9}$	$\frac{1}{3}$	1	3
g(x)	$\frac{1}{16}$	$\frac{1}{4}$	1	4



Department of Education General Mathematics Learner's Material

The two graphs have the same domain, range, y-intercept, and horizontal asymptote. However, the graph of g(x) = 4^x rises faster than does f(x) = 3^x as x increases, and is closer to the x-axis if x < 0

For both functions:

Domain: Set of all real numbers

Range: Set of all positive real numbers

y-intercept: (0,1). There is no x-intercept

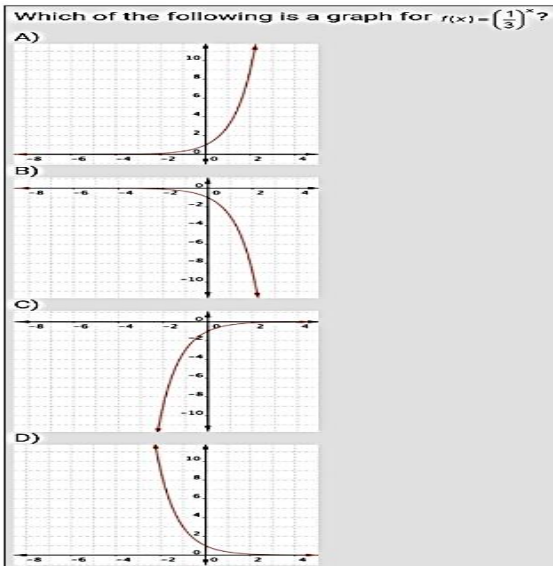
Horizontal Asymptote: y = 0

Let's do this:

1. Construct a table of values for the given function f(x) = 5^x using
   a. the values of x = -2, -1, 0, 1, 2.
   b. Sketch their graphs on a coordinate plane.
   c. For each function, label the domain, range, y-intercept and horizontal asymptote.

Assessment

1.



2. Construct a table of values for the given functions below using
(a) the values of
x = -2, -1, 0, 1, 2.
(b) Sketch their graphs on a coordinate plane.
(c) For each function, label the domain, range, y-intercept and horizontal asymptote.
a. f(x) = $(\frac{1}{5})^x$
b. f(x) = $(6)^x$
c. f(x) = $(3)^{-x}$

Additional Activities

Graph each exponential function and identify the domain and range.

1. f(x) =10^x
   domain: _____
   range: _____
2. g(x) = $(\frac{1}{10})^x$
   domain: _____
   range: _____

Lesson 7	Solving Exponential Equations and Inequalities

What I Need to Know

After going through this module, you are expected to solve problems involving exponential functions, equations, and inequalities (M11GM-Ig-2).

What I Know

Choose the letter of the correct answer. Write the chosen letter on a separate sheet of paper.

_____1.) Which of the following is an example of exponential inequality?

A. $4^{x-1} = 2^{x+5}$
B. $y = 3(2)^x$
C. f(x) = $8(\frac{3}{4})^x$
D. $5^{2x-1} > \frac{1}{125}$

_____2.) What is the value of x in an equation $9^{x+1} = 27^{x+4}$

A. x = 10
B. x = -10
C. x = -5
D. x = 5

_____3.) In an equation $36^{2x+4} = 216^{x+5}$, solve for x?

A. x = 5
B. x = 6
C. x = 7
D. x = 8

_____4.) Which of these is an exponential function?

A. $f(x) = 9^{x+4}$
B. $2^{3x} = 4^{4x+6}$
C. $25^{2x+2} < 125^{3x+1}$
D. $\frac{1}{16} < 4^{2x}$

_____5.) Which of the following is an equation involving exponential expression?

A. Exponential inequality
B. Exponential function
C. Exponential equation
D. None of these

What's In

Before we begin graphing, it is helpful to review the behavior of exponential growth. Recall the table of values for a function of the form f(x) = b^x whose base is greater than one. We'll use the function f(x) = 2^x Observe how the output values in the table below change as the input increases by 1.

Definition: An exponential expression is an expression of the form $\boldsymbol{a \,.\, b^{\,x-c} + d}$**,** where
(b > 0, b ≠ 1).

The definitions of exponential equations, inequalities and functions are shown below.

	Exponential Equation	Exponential Inequality	Exponential Function
Definition	An equation involving exponential expressions	An inequality involving exponential expressions	Function of the form f(x) = b^x**,** where (b > 0, b ≠ 1).
Example	$7^{2x-x^2} = \frac{1}{343}$	5^{2x} - 5^{x+1} ≤ 0	f(x)= $(1.8)^x$ or y= $(1.8)^x$

What's New

One-to-one Property of Exponential Functions

If $x_1 \neq x_2$ *then* $b^{x_1} \neq b^{x_2}$. *Conversely, if* $b^{x_1} = b^{x_2}$ *then* $x_1 = x_2$

What is It

Example 1. Solve the equation $4^{x-1} = 16$.

Solution. Write both sides with 4 as the base.

$4^{x-1} = 16$
$4^{x-1} = 4^2$
x -1 =2
x = 2 + 1
x = 3

Alternative Solution.
Write both sides with 2 as the base

$4^{x-1} = 16$
$2^{2(x-1)} = 2^4$
2(x -1) = 4
2x - 2 = 4
2x = 6
x = 3

Example 2. Solve the equation $125^{x-1} = 25^{x+3}$

Solution. Both 125 and 25 can be written using 5 as the base.

$125^{x-1} = 25^{x+3}$
$5^{3(x-1)} = 5^{2(x+3)}$
3(x-1) = 2(x+3)
3x -3 = 2x + 6
x = 9

What's More

Example 3. Solve the inequality $3^x < 9^{x-2}$
Solution. Both 9 and 3 can be written using 3 as the base.

$3^x < 3^{2(x-2)}$
$3^x < 3^{2x-4}$

Since the base 3 > 1, then this inequality is equivalent to
x < 2x – 4 (the direction of the inequality is retained)
4 < 2x – x
4 < x
The solution set to the inequality is $\{ x \in \mathbb{R} \mid x > 4 \}$.

What I Have Learned

Solve for x in the following equations or inequalities.

1. $3^x = 81$
Solution.

$3^x = 3^4$
x = 4

Make the equation be the same base

2. $(\frac{4}{6})^x \geq \frac{34}{16}$

Solution.

$(\frac{4}{6})^x \geq \frac{6^2}{4^2}$ Simplify

$(\frac{4}{6})^x \geq (\frac{6}{4})^2$

$(\frac{4}{6})^x \geq (\frac{4}{6})^{(-1)^2}$ Get the reciprocal to make the base the same by multiplying -1

$(\frac{4}{6})^x \geq (\frac{4}{6})^{-2}$

The solution set is $\{ x \in \mathbb{R} \mid x \leq -2 \}$

What I Can Do

Let's do this: Solve for x.

1. $169^x = 13^x$
2. $7^x = \frac{1}{49}$
3. $(\frac{3}{5})^{x+1} = \frac{25}{9}$
4. $4^{3x+2} < 64$
5. $(\frac{49}{81})^{x+1} \geq \frac{9}{7}$
6. $4^{x+1} = \frac{1}{64}$
7. $5^{3x+8} = 25^{2x}$
8. $4^{5x-13} = \frac{1}{8^x}$
9. $10^x > 100^{-2x-5}$
10. $49^x = 343^{2x-3}$

Assessment

Solve each equation.

1. $2^{5x} = 4^{2x-1}$
2. $(\frac{1}{2})^{4x+1} = 8^{2x+1}$
3. $7^{3x-1} = 49^{x+1}$
4. $(\frac{2}{3})^{5x+1} = (\frac{9}{4})^{x-4}$
5. $4^{2x+6} \leq 64^{2x-4}$
6. $25^{y-3} \leq \frac{1}{125}^{y+2}$
7. If a chemist has **40** grams of a substance that has a **half-life** of **6** hours, how much will there be after **18** hours?

Additional Activities

1. Solve each of the following problem.
   a. $5^{3x} = 5^{7x-2}$
   b. $4^{t^2} = 4^{6-t}$
   c. $4^{5-9x} = \frac{1}{8^{x-2}}$
   d. $16^{x-1} > 2^{2x+2}$
   e. $5^{x+3} \leq 25^{x+4}$

2. Megan has **₱20,000** to invest for **5** years and she found an interest rate of **5%**. How much money will she have at the end of **5** years if:
(a) the interest rate compounds monthly?
(b) the interest rate compounds semi-annually?

Lesson 8	Real-life situations using logarithmic functions

What I Need to Know

After going through this module, you are expected to:

1. represent real-life situations using logarithmic functions (M11GM-Ih-1);
2. distinguish logarithmic function, logarithmic equation, and logarithmic inequality (M11GM-Ih-2);
3. solve logarithmic equations and inequalities (M11GM-Ih-i-1).

What I Know

Choose the letter of the correct answer. Write the chosen letter on a separate sheet of paper.

_____1.) What property of logarithmic function is, If b is a positive number other than 1 then the $log_b x = log_b y$ if and only if x = y?

A. Equality C. Function
B. Expression D. Inequality

_____2.) It is a logarithm to the base e (approximately 2.71828)?

A. Common logarithm C. Logarithmic function
B. Logarithmic expression D. Natural logarithm

_____3.) What types of logarithms is in the form f(x) =$log_b x$ (b > 0, b ≠ 1)?

A. Common logarithms C. Logarithmic functions
B. Logarithmic expressions D. Natural logarithms

_____4.) A property of logarithmic functions If b > 1, then log_b x > log_b y if and only if x > y and log_b x < log_b y if and only if x < y.?

A. Equality C. Function
B. Expression D. Inequality

_____5.) These are logarithms with base 10; log x is a short notation for $log_{10}x$?

A. Common logarithms C. Logarithmic functions
B. Logarithmic expressions D. Natural logarithm

What's In

Definition: Let a, b, and c be positive real numbers such that b ≠ 1. The logarithm of a with base b is denoted by $log_b a$ and is defined as **c = $log_b a$ if and only if a = b^c**

Common logarithms are logarithms with base 10; log x is a short notation for $log_{10}x$.

Natural logarithms are logarithms to the base e (approximately 2.71828), and are denoted by **"ln"**. In other words, ln x is another way of writing for $log_e x$.

Reminders.

1. In both the logarithmic and exponential forms, b is the base. In the exponential form, c is an exponent; this implies that the logarithm is actually an exponent. Hence, logarithmic and exponential functions are inverses.

2. In the logarithmic form $\log_b x$, x and b cannot be negative.

What's New

Example 1. Rewrite the following exponential equations in logarithmic form, whenever possible.

a.$5^3 = 125$ b. $7^{-2} = \frac{1}{49}$ c. $(\frac{2}{3})^2 = \frac{4}{9}$ d. $10^2 = 100$ e. $(0.1)^{-4} = 1000$

f. $4^0 = 1$ g. $7^b = 21$ h. $e^2 = x$ i. $(-2)^2 = 4$

Solution.

a.$log_5 125 = 3$ b. $log_7 \frac{1}{49} = -2$ c. log 100 = 2 d. $log_{\frac{2}{3}}(\frac{4}{9}) = 2$ e. $log_{0.1}1000 = -4$

f. $log_4 1 = 0$ g. $log_7 21 = b$ h. lnx = 2 i. cannot be written in logarithmic form

Example 2. Rewrite the following logarithmic equations in exponential form.

a.log m= n b. $log_3 81 = 4$ c. $log_{\sqrt{5}} 5 = 2$ d. $log_{\frac{3}{4}}(\frac{64}{27}) = -3$

e. $log_4 2 = \frac{1}{2}$ f. $log_{10} 0.001 = -3$ g. ln8 = a

Solution.

a.$10^n = m$ b. $3^4 = 81$ c. $(\sqrt{5})^2 = 5$ d. $(\frac{3}{4})^{-3} = \frac{64}{27}$

e. $4^{\frac{1}{2}} = 2$ f. $10^{-3} = 0.001$ g. $e^a = 8$

What is It

	Logarithmic Equation	Logarithmic Inequality	Logarithmic Function
Definition	An equation involving logarithms.	An inequality involving logarithms.	Function of the form f(x) =$log_b x$ (b > 0, b ≠ 1).
Example	$log_x 2 = 4$	$\ln x^2 > (\ln x)^2$	g(x) = $log_3 x$

Logarithmic Equations

a.log_2 2x = 3

log_2 2x = 3	Original equation
2x = 2^3	Definition of logarithm
2x = 8	$2^3 = 8$
x = 4	Simplify

b. log_2 (x + 17) = log_2 (3x + 23)

Since the bases of the logarithms are equal, (x + 17) must equal (3x + 23).

(x + 17) = (3x + 23)	Property of equality
-6 = 2x	Simplify
x = -3	Divide each side by 2.

What's More

log_3(x^2+ 5) = 2

log_3(x^2+ 5) = 2	Original equation
(x^2+ 5) = 3^2	Definition of logarithm
x^2 + 5 = 9	Simplify.
x^2= 4	Subtract 5 from each side
x = ±2	Simplify.

Logarithmic Inequality

a.log_5 (4x - 3) < 3

log_5 (4x - 3) < 3	Original inequality
0 < 4x - 3 <5^3	Property of Inequality for Logarithmic Functions

$3 < 4x < 125 + 3$	Simplify.
$\frac{3}{4} < x < 32$	Simplify.

The solution set is $\{x \mid \frac{3}{4} < x < 32\}$.

b. log_3 (3x - 4) < log_3 (x + 1)

$log_3 (3x - 4) < log_3 (x + 1)$	Original inequality
$3x - 4 < x + 1$	Property of Inequality
$2x - 4 < 1$	Subtract x from each side.
$2x < 5$	Add 4 to each side
$x < \frac{5}{2}$	Divide each side by 2.

3x - 4 and x + 1 must both be positive numbers. Use this information to constrain your solution.

$3x - 4 > 0$	$x + 1 > 0$
$3x > 4$	$x > -1$
$x > \frac{4}{3}$	**The solution is $\{x \mid \frac{4}{3} < x < \frac{5}{2}\}$.**

These properties also hold true for ≤ and ≥. When solving logarithmic inequalities, make sure you identify extraneous solutions. If a solution results in taking a logarithm of a number less than or equal to zero, it is extraneous.

What I Have Learned

<u>Earthquake Magnitude on a Richter scale</u>

The magnitude R of an earthquake is given by $R = \frac{2}{3}\log\frac{E}{10^{4.40}}$ where E (in joules) is the energy released by the earthquake (the quantity $10^{4.40}$ joules is the energy released by a very small reference earthquake).The formula indicates that the magnitude of an earthquake is based on the logarithm of the ratio between the energy it releases and the energy released by the reference earthquake.

Example 1. Suppose an earthquake released approximately10^{12}energy.
(a) What is its magnitude on a Richter scale?
(b) How much more energy did this earthquake release than the reference earthquake?

Solution.

(a) Since $E = 10^{12}$, then $R = \frac{2}{3}\log\frac{10^{12}}{10^{4.40}} = \frac{2}{3}log10^{7.6}$ Since by definition, log $10^{7.6}$ is the exponent by which 10 must be raised to obtain $10^{7.6}$, then $\log 10^{7.6} = 7.6$

Thus, $R = \frac{2}{3}(7.6) \approx 5.1$

(b) This earthquake releases $\frac{10^{12}}{10^{4.40}} = 10^{7.6} \approx 39810717$ times more energy than the reference earthquake.

<u>Sound Intensity</u>

In acoustics, the decibel (dB) level of a sound is $D = 10 \log \frac{I}{10^{-12}}$

where **I** is the sound intensity in watts/m^2(the quantity 10–12 watts/m^2 is least audible sound a human can hear).

Example 2. The decibel level of sound in a quiet office is 10log 10^{-6} watts/m^2 (a) What is the corresponding sound intensity in decibels? (b) How much more intense is this sound than the least audible sound a human can hear?

Solution.

a) D = 10 log $\frac{I}{10^{-12}}$ = 10log 10^{-6} Since by definition, log10^6 is the exponent by which 10 must be raised to obtain 10^6 then log 10^6 = 6

Thus, D = 10(6) = 60 decibels.

b) This sound is $\frac{10^{-6}}{10^{-12}} = 10^6$ = 1,000,000 times more intense than the least audible sound a human can hear.

<u>Acidity and the pH scale</u>

The pH level of a water-based solution is defined as $pH = -\log[H^+]$

Where $[H^+]$ is the concentration of hydrogen ions in moles per liter. Solutions with a pH of 7 are defined **neutral**; those with pH < 7 are **acidic**, and those with pH > 7 are **basic**.

What I Can Do

Determine whether the given is a logarithmic function, a logarithmic equation, a logarithmic inequality or neither

1. $g(x) = log_5$ x
2. $y = 2log_4$ x
3. log$(4x) = -log$ $(3x + 5)$
4. $xlog_2(x)$ - 1> 0
5. log x(x-3) = log 4

Assessment

A. Determine whether the given is a logarithmic function, a logarithmic equation, a logarithmic inequality or neither.

1. $log_3(2x - 1) = log_3 x + 2$
2. $log_3(2x - 1) = 2$
3. $h(x) = log_{0.25}\, x$
4. $2 + y = log_3 x$
5. $\log x^2 = 2$

B. In numbers 1-3, find the value of the following logarithmic expressions.

1. $log_2 \frac{1}{2}$ 2. $log_{100}\, 10$ 3. $log_{0.5}\, 4$

Additional Activities

Solve this problem: What is the magnitude in the Richter scale of an earthquake that released 10^{14} joules of energy? How much more energy does this earthquake release than that of the reference earthquake?

Lesson 9	Graphing Logarithmic Functions

What I Need to Know

After going through this module, you are expected to:

1. represent a logarithmic function through its: (a) table of values, (b) graph, and (c) equation (M11GM-Ii-2);
2. find the domain and range of a logarithmic function (M11GM-Ii-3);
3. determine the intercepts, zeroes, and asymptotes of logarithmic functions (M11GM-Ii-4);
4. solve problems involving logarithmic functions, equations, and inequalities (M11GM-Ij-2).

What I Know

Choose the letter of the correct answer. Write the chosen letter on a separate sheet of paper.

_____1.) If the domain of f(x) = $log_b x$ is (0, ∞), what is its range?

A. $(-\infty, \infty)$ C. $(\infty, -\infty)$
B. (∞, ∞) D. $(-\infty, -\infty)$

_____2.) If the x-intercept of f(x) = $log_b x$ is (1, 0), what is its y-intercept?

A. (0, 1) C. (0, −1)
B. (-1, 0) D. None

_____3.) What is the vertical asymptote of the logarithmic function is?

A. x= 0 C. x=3
B. x=1 D. No vertical asymptote

_____4.) The graph of logarithmic function is increasing if?

A. b > 1 C. b ≤ 1
B. b < 1 D. b ≠ 1

_____5.) The graph of logarithmic function is decreasing if?

A. 0 > b < 1 C. 0 < b < 1
B. 0 < b > 1 D. None of these

What's In

We begin with the parent function $y = log_b\, x$, Because every logarithmic function of this form is the inverse of an exponential function with $y = b^x$ the form their graphs will be reflections of each other across the line y=x. To illustrate this, we can observe the relationship between the input and output values of y = 2^x and its equivalent $x = log_2$ (y) in the table below.

REMEMBER… For x > 0 and b > 0, and b = 1 So…x and b MUST be positive and b can never equal to 1 (this is why we have asymptotes in our graphs)
To Solve a Logarithm with a log on each side…
1. Make sure the log and the base are IDENTICAL.
2. If the logs are identical, then the numbers within in the log are identical so set them equal.
3. Solve for x…YOU MAY HAVE TO FACTOR!
4. Check Solution to be sure x is positive!

What's New

A GENERAL NOTE: CHARACTERISTICS OF THE GRAPH OF THE PARENT FUNCTION, $F(x) = log_b\, x$

For any real number x and constant $b > 0$, $b \neq 1$, we can see the following characteristics in the graph of $F(x) = log_b\, x$

- one-to-one function
- vertical asymptote: $x = 0$
- domain: $(0,\infty)$
- range: $(-\infty,\infty)$
- x-intercept: (1,0) and key point $(b,1)$
- y-intercept: none
- increasing if $b>1$

Property of Logarithmic Equations

If b > 1, then the logarithmic function y = $log_b\, x$ is increasing for all x. If 0 < b < 1, then the logarithmic function y = $log_b\, x$ is decreasing for all x. This means that $log_b\, u = log_b\, y$ if and only if u = v.

1.Let's finish solving the problem $log_3(7x + 3) = log_3\,(5x + 9)$

$log_3(7x + 3) = log_3\,(5x + 9)$

Since the problem has only two logarithms on opposite sides of the equal sign, the problem can be solved by dropping the logarithms.

7x + 3 = 5x + 9

Drop the logarithms.

2x = 6

Finish solving by subtracting 5x from each side, subtracting 3 from each side, and finally dividing each side by 2.

x = 3

Check the answer; this is an acceptable answer because we get a positive number when it is plugged back in.

Therefore, the solution to the problem $log_3(7x + 3) = log_3\,(5x + 9)\ is\ x = 3$

2. Here is another example, solve $log_7(x - 2) = log_7\,(x + 3)$

$log_7(x - 2) = log_7\,(x + 3)$

This problem can be simplified by using Property 3 which changes the addition of logarithms to multiplication.

(x - 2)(x + 3) =14

Drop the logarithms. Then, simplify the problem by distributing or FOILing and combining like terms.

$x^2 - x - 6 = 14$

$x^2 - x - 20 = 0$

Solve the problem by subtracting 14 from each to get it equal to zero, and then factoring or using the quadratic formula to find the values of x.

(x + 4)(x - 5) = 0

x = - 4 or x = 5

Check the answers, only one answer is acceptable because the other answer produces a negative number when it is plugged back in.

Therefore, the solution to the problem $log_7(x-2) = log_7\ (x+3)$ is x = 5.

What is It

PROPERTIES OF LOGARITHMIC FUNCTIONS:
1. The domain is the set of all positive numbers, or
2. The range is the set of all positive real numbers.
3. It is a one-to-one function. It satisfies the Horizontal Line Test.
4. The x-intercept is (1,0). There is no y-intercept.
5. The vertical asymptote is the line x = 0 (or the y-axis). There is no horizontal asymptote.

Properties of Logarithmic Inequalities

If 0 < b < 1, then $x_1 < x_2$ if and only if $log_b x_1\ > log_b x_2$.

If b > 1, then $x_1 < x_2$ if and only if $log_b x_1\ < log_b x_2$.

3. Solve the following logarithmic inequalities.

a. $\boldsymbol{log_3\ (2x-1) >\ log_3\ (x+2)}$

Solution.

Step 1: Ensure that the logarithms are defined.

Then 2x – 1 > 0 and x + 2 > 0 must be satisfied.

2x – 1 > 0 implies x > $\frac{1}{2}$ and x + 2 > 0 implies x > –2.

To make both logarithms defined, then x > $\frac{1}{2}$. (If x > $\frac{1}{2}$, then x is surely greater than –2.)

Step 2: Ensure that the inequality is satisfied.

The base 3 is greater than 1.

Thus, since $Log_3\ (2x-1) >\ Log_3\ (x+2)$ then:

2x – 1 > x + 2

x > 3 (subtract x from both sides; add 1 to both sides)

x > 3

Hence, the solution is (3, +∞).

b. –2 < logx < 2

Solution.

Step 1: Ensure that the logarithms are defined.

This means that x > 0.

Step 2: Ensure that the inequality is satisfied.

Rewrite –2 and 2 as logarithms to the base 10, which are $\log 10^{-2}\ and\ \log 10^{2}$ respectively, obtaining the inequality: $\log 10^{-2} < \log x <\ \log 10^{2}$

We split the compound inequality into two simple inequalities:

$\log 10^{-2} < \log x \; and \; \log x < \log 10^{2}$

Since the base 10 is greater than 1, simplify both inequalities as $10^{-2} < x < 10^{2}$

Thus obtaining 1/100 < x < 100, which automatically satisfies the condition in Step 1.

Hence, the solution is $(\frac{1}{100}, 100)$.

What's More

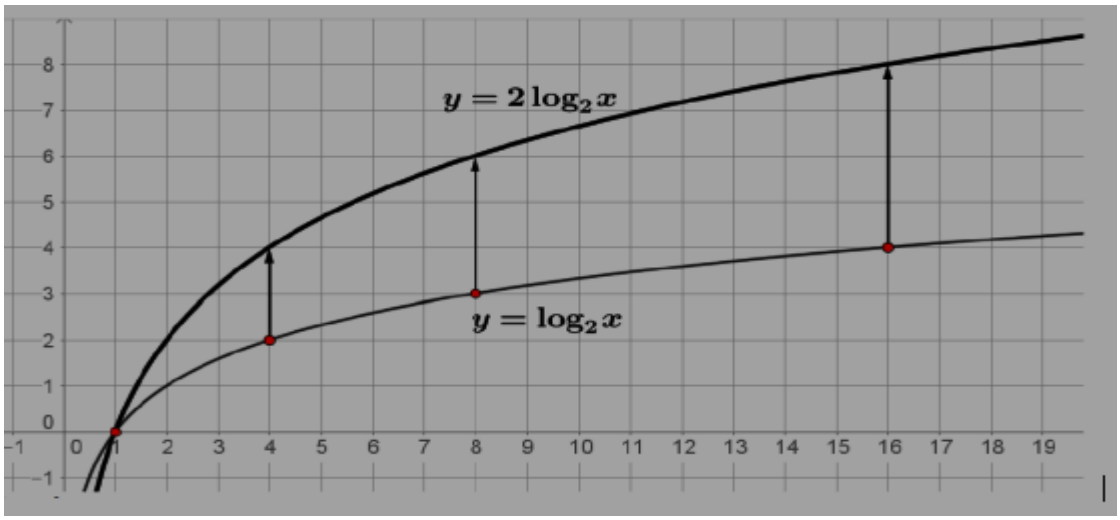
Example 1. Sketch the graphs of $y = 2log_2 x$. Determine the domain, range, vertical asymptote, x-intercept, and zero.

Solution.

The graph of $y = 2log_2 x$ can be obtained from the graph of $y = 2log_2 x$ by multiplying each y-coordinate by 2, as the following table of signs shows.

X	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8
$log_2 x$	-4	-3	-2	-1	0	1	2	3
$y = 2log_2 x$	-8	-6	-4	-2	0	2	4	6

The graph is shown below.



Analysis:

a. Domain: $\{x \mid x \in \mathbb{R}, x > 0\}$

b. Range: $\{y \mid y \in \mathbb{R}\}$

c. Vertical Asymptote: x = 0

d. x-intercept: (1, 0)

e. Zero: 1

What I Have Learned

Steps for Solving Logarithmic Equations Containing Terms without Logarithms

Step 1: Determine if the problem contains only logarithms. If so, stop and use Steps for Solving Logarithmic Equations Containing Only Logarithms. If not, go to Step 2.

Step 2: Use the properties of logarithms to simplify the problem if needed. If the problem has more than one logarithm on either side of the equal sign then the problem can be simplified.

Step 3: Rewrite the problem in exponential form.

Steps for Solving Logarithmic Equations Containing Terms without Logarithms

Step 4: Simplify the problem.

Step 5: Solve for x.

Step 6: Check your answer(s). Remember we cannot take the logarithm of a negative number, so we need to make sur DO_Q1_GenMathSHS_Lesson9 back into the original equation we get a positive number. Otherwise, we must drop that answer(s).

Solving the problem $log_2\,(5x+7) = 5$

$log_2\,(5x+7) = 5$	This problem does not need to be simplified because there is only one logarithm in the problem.
$5x + 7 = 2^5$	Rewrite the problem in exponential form by moving the base of the logarithm to the other side.
$5x + 7 = 32$	Simplify the problem by raising 2 to the fifth power.
$x = 5$	Solve for x by subtracting 7 from each side and then dividing each side by 5.
$x = 5$	Check the answer; this is an acceptable answer because we get a positive number when it is plugged back in.

Therefore, the solution to the problem $log_2\,(5x+7) = 5$ is x = 5.

What I Can Do

Let's do this:

A. Analyze each of the following functions by:
   a) using the transformations to describe how the graph is related to a logarithmic function $y = log_b\,x$,
   b) identifying the x-intercept, vertical asymptote, domain and range, and
   c) sketching the graph of the function.

1. $f(x) = log_2(x-3)$
2. $g(x) = log_{0.5}(x) - 3$

B. Find the value/s of x of the following logarithmic equations/inequalities.

1. log 15x = Log 30
2. log 3x-2 = Log 2
3. $\log_x 121 = 2$
4. $(log_5 x)^2 + 5log_5 x + 6$
5. $\log_3 (9x) - \log_3 (8 - x) = 4$
6. $\log_8 (3x - 5) < 2$
7. $\log_4 (x + 1) < \log_4 (2x)$
8. $-5 < \log x < 5$

Assessment

A. Sketch the graph and identify the domain and range.

1. $f(x) = log_6(x - 1) - 5$
2. $s(x) = log_5(x - 1) + 3$
3. $y = log_6(x - 3) - 2$

B. Solve the following logarithmic equations/inequalities.

1. $log_3 (x + 4) = log_3 (2x - 4)$
2. $log_2 (3x + 5) \geq log_2 (x - 9)$
3. $log_2(x)^2 - 4 = 0$
4. $\log x + \log (x + 3) \leq \log 10$

Additional Activities

Construct a table of values for the given functions below using

(a) the values of x = -2, -1, 0, 1, 2.

(b) Sketch their graphs on a coordinate plane.

(c) For each function, identify the domain, range, y-intercept and vertical asymptote.

1. f(x) = log_4 (x-2) + 1
2. f(x) = log_3 (x-2) - 2
3. f(x) = log_2 (x+6) + 1

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B. Electronic Resources

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https://courses.lumenlearning.com/wmopencollegealgebra/chapter/introduction-graphs-of-exponential-functions/

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